

Carl A. Ziegler, Ph.D.

Exoplanet Astronomer | Associate Professor (August 2026) | Director, SFA Observatory
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Academic Profile

Observational astrophysicist and university educator with more than a decade of experience in exoplanet characterization, stellar multiplicity, adaptive optics, speckle interferometry, and robotic surveys. Leads international high-resolution imaging programs for Kepler and TESS, develops automated Python analysis pipelines, and translates research into instrumentation, open-source software, authentic student research, and public astronomy programs.

180+ refereed publications | 5,400+ career citations | h-index 49 (Google Scholar) | 4,800+ Kepler/TESS hosts imaged | 20+ student researchers mentored

Research and Technical Expertise

Research: exoplanet detection and characterization; planetary systems in multiple stars; high-angular-resolution imaging; survey and instrument design; selection effects, uncertainty analysis, Bayesian modeling, Monte Carlo simulation, time-series photometry, and population inference.

Technical: Python, Astropy, emcee, JavaScript, HTML/CSS, C++, LabVIEW, Mathematica, Zemax, and SolidWorks; automated reduction pipelines, image registration, PSF subtraction, astrometry, contrast curves, robotic telescope control, scientific visualization, and browser-based simulation.

Academic Appointments

Assistant Professor of Astronomy; Associate Professor effective Aug. 2026

2020–Present

Stephen F. Austin State University

Nacogdoches, TX

- Director of the SFA Observatory, leading research, instructional, and public-outreach operations, student training, observing programs, eclipse events, and regional partnerships.
- Teach and develop courses from introductory astronomy through astrophysics, observational astronomy, mathematical methods, and senior practicum; design exams, laboratories, projects, and rubrics aligned with explicit learning objectives.
- Mentored more than 20 undergraduate and graduate researchers in Python, photometry, high-resolution imaging, exoplanet follow-up, and scientific software, resulting in refereed coauthorships, conference presentations, REU placements, graduate study, and technical careers.

Dunlap Postdoctoral Fellow

2018–2020

Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Toronto, Canada

- Founded the SOAR TESS Survey and One-Hit Wonders program; developed automated target selection, observing, reduction, and validation workflows for high-resolution imaging and long-period transiting planets.

Research Leadership and Scholarly Innovation

SOAR TESS Survey, Principal Investigator (2018–Present). Lead a systematic speckle-imaging survey on the 4.1-m SOAR Telescope using thousands of observations of TESS Objects of Interest. The survey has resolved companions to more than 1,000 targets and demonstrated that close binary systems suppress detected planets by nearly a factor of seven; uncorrected blending can bias inferred planet occurrence rates by approximately a factor of two. Coordinate telescope operations, target prioritization, collaborators, student contributors, data products, and publication cycles.

Robo-AO Kepler Survey and Robo-AO-2. Collaborators and I obtained high-resolution images of all 3,857 Kepler planet-candidate systems, identifying blends and false positives, revising hundreds of planet radii, and supporting validation of more than 1,500 Kepler and K2 planets. Co-designed a proposed dual-band visible/near-infrared survey of approximately 4,000 TESS targets with public delivery through ExoFOP.

Gravitas, Creator and Lead Developer (2025–Present). Designed and deployed an open-source browser-based astronomy simulation, co-developed with an undergraduate researcher, for orbital dynamics, black holes, energy conservation, spacetime, gravitational-wave sonification, and exoplanet transit light curves. Integrated it into guided-inquiry instruction. [Launch Gravitas](#).

Teaching, Mentorship, and Outreach

- **Documented teaching effectiveness:** 2025 annual student-evaluation average of 4.85/5; evaluations regularly exceeded 4.5/5 across ten semesters. Formal peer and administrative reviews document sustained growth and pedagogical innovation.
- Rebuilt Observational Astronomy as a semester-long authentic research experience and developed an upper-level Astrophysics II course with original notes, problem sets, examinations, and worked solutions; completed four ACUE courses and SFA Certified Online Instructor training.
- Organized more than 20 public observing nights since 2021, reaching more than 2,000 visitors; led a 2023 annular-eclipse event for more than 200 attendees and supported 2024 total-eclipse programming, school visits, Scout events, camps, and regional STEM partnerships.

Selected Publications

- Ziegler, C., et al. (2021), “SOAR TESS Survey II: The Impact of Stellar Companions on Planetary Populations,” *AJ*, 162, 192.
- Ziegler, C., et al. (2020), “SOAR TESS Survey I: Sculpting of TESS Planetary Systems by Stellar Companions,” *AJ*, 159, 19.
- Ziegler, C., et al. (2018), “Measuring the Recoverability of Close Binaries in Gaia DR2 with the Robo-AO Kepler Survey,” *AJ*, 156, 259.
- Ziegler, C., et al. (2018), “Robo-AO Kepler Survey IV: The Effect of Nearby Stars on 3,857 Planetary Candidate Systems,” *AJ*, 155, 161.

Education

Ph.D., Physics, 2018, University of North Carolina at Chapel Hill. Dissertation: *Characterization of Exoplanets and Stellar Systems with New Robots*. **M.S., Physics, 2013**, Southern Illinois University Carbondale. **B.A., Physics and Mathematics, 2009**, William Jewell College.